

Haoran (Alex) Yu

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SUMMARY

- Adaptable, collaborative, and detail-oriented professional with a strong mathematical foundation and problem-solving
- Proficient in Python libraries such as NumPy, pandas, Matplotlib, scikit-learn, and SciPy for data manipulation, visualization, scientific computing, and machine learning
- Experiences in using DEAP (Distributed Evolutionary Algorithms in Python) for optimization with evolutionary algorithm
- Proficient in R and core tidyverse packages
- Experiences in using SAS for data management, and SQL for querying, manipulating, and analyzing data from relational databases
- Proficient in LaTeX

EDUCATION

HBSc., Major in Mathematics, Minors in Statistics and Economics

09/2021 – 07/2025 (Exp')

University of Toronto St. George Campus

Toronto, ON

Relevant Courses: Probability and Statistics, Financial Economics, Methods of Data Analysis, Statistical Machine Learning, Linear Algebra, Advanced Calculus, Abstract Mathematics, Real Analysis, Complex Analysis, Mathematical Modelling, Nonlinear Optimization, Mathematical Theory of Finance

PROJECT EXPERIENCE

Machine Learning Perspective on Predicting Loan Default Risks in Financial Services

08/2024 - 12/2024

- Acquired data preprocessing skills by cleaning, imputing, and encoding features from LendingClub loan dataset
- Strengthened exploratory analysis abilities by examining distributions and correlations to pinpoint default predictors
- Developed modeling expertise by training a balanced Random Forest and tuning key hyperparameters
- Gained evaluation experience by interpreting confusion matrices, classification reports, and ROC/AUC scores
- Improved documentation skills by narrating the workflow and visualizing results in a Jupyter Notebook

Credit Card Fraud Detection With Machine Learning

10/2024 – 1/2025

- Acquired data preprocessing skills by cleaning and imputing features from 2023 European cardholders fraud dataset from Kaggle
- Developed modeling expertise by implementing and fine-tuning classifiers (Logistic Regression, Random Forest, XGBoost) using cross-validation
- Gained evaluation experience by interpreting confusion matrices, precision-recall curves, and ROC/AUC scores
- Improved documentation skills by narrating the end-to-end workflow and embedding code, results, and visualizations in a Jupyter Notebook

Regression Analysis on Analysis Premium

07/2024 - 08/2024

- Cultivated collaboration skills by working in a group to predict insurance premiums using real-life data
- Strengthened data preparation abilities by applying data selection, cleaning, and transformation techniques
- Gained expertise in model construction and refinement by building a linear regression model using Box-Cox transformations, square root, log, and Ridge regression
- Developed proficiency in model evaluation by assessing performance using F-statistics, partial F-statistics, Cook's distance, and R-squared
- Enhanced communication skills through summarizing findings and conclusions in a comprehensive project report

WORK EXPERIENCE

Real Estate Investment Expansion Intern Analyst

06/2023 – 09/2023

Shaanxi Jintai Hengye Real Estate Co., Ltd.

Xi'an, China

- Participated in the evaluation of land acquisitions for projects, including public auctions and equity acquisitions, analyzing land value and potential returns
- Conducted real estate market research, collecting and analyzing data on competitors, market trends, and consumer demands to provide data-driven insights for project decision-making
- Developed and compiled project feasibility study reports, covering market analysis, SWOT analysis, preliminary planning, and financial forecasting, enhancing analytical and report writing skills